

User Manual

Bike Crash Detection System

1. Introduction

The Bike Crash Detection System is designed to automatically detect accidents using machine learning and sensor data. It helps provide faster emergency response by identifying crash events.

2. Requirements

Hardware Requirements

- Smartphone or sensor device
- Accelerometer
- Gyroscope
- GPS

Software Requirements

- Python
- TensorFlow
- Keras
- Jupyter Notebook / IDE

3. Installation Steps

1. Install Python.
2. Install required libraries:

```
pip install tensorflow  
pip install numpy  
pip install pandas
```

1. Download the project source code.
2. Open the project folder.

4. Running the Project

1. Open the project in Python IDE/Jupyter Notebook.
2. Load the required dataset.
3. Run preprocessing module.
4. Train/load the CNN-LSTM model.
5. Start crash detection.

5. Working Process

1. Sensors collect bike movement data.
2. Data is processed.
3. CNN extracts features.
4. LSTM analyzes motion sequence.
5. System predicts:
 - Crash Detected
 - Normal Riding
6. If crash detected, alert is generated.

6. Output

The system displays:

- Crash status
- Detection result
- Emergency alert information

7. Troubleshooting

Problem: Model not running

Solution: Check Python libraries.

Problem: Sensor data unavailable

Solution: Check sensor connection.

8. Future Improvements

- Mobile application integration
- IoT implementation
- Cloud monitoring
- More sensor integration